

Review of the potential for recycling CO₂ from organic waste composting into plant production under controlled environment agriculture

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ABSTRACT

Composting has been used extensively as a sustainable approach for the management organic waste streams and conversion into stabilized soil amendments and horticultural growth media. In addition to these products, the composting process also generates by-products of microbial metabolism that includes gas (CO₂) and energy (heat). High rates of food waste and commercial organic waste generation, along with greater diversion away from landfills, creates new opportunities to valorize these by-products from aerobic composting. Specifically, integrating carbon dioxide recapture and recycling into Controlled Environment Agriculture (CEA) production systems presents a pathway to circularize resources. CEA systems produce plants year round in an environmentally controlled space where elevating CO₂ environments can boost plant production, creating a synergistic relationship with composting processes. The review evaluated CO₂ emissions from an array of open and closed composting systems, as well as different feedstocks to demonstrate its potential for use in CEA. Rates of CO₂ emissions varied widely, with a peak CO₂ emission range of 0.32–429.27 g CO₂-C kg⁻¹ d⁻¹ depending on the aeration conditions and feedstock composition. Analysis of two hypothetical CEA scenarios reveals that integrating in-vessel composting as an on-site waste management process is comparable in cost to using natural gas as a regular CO₂ source. The quantity of composting feedstocks required to generate CO₂ emissions for a single tier and five tier CEA lettuce production system ranged from 3.38 to 69.94 kg dry biomass d⁻¹ and 9.93–205.50 kg dry biomass d⁻¹, respectively, but varied based on feedstock composition and compost operating conditions. The long-term research goal is toward valorizing non-edible biomass of CEA systems to create a circular economy model of commercial plant production.

1. Introduction

Rapid urbanization aligned with global population growth results in significant quantities of organic wastes being generated annually (Bong et al., 2017; Karak et al., 2012; Narayana, 2009). These include food wastes from residential, commercial, and manufacturing sectors (Melikoglu et al., 2013; Julian et al., 2010) as well as secondary food processing by-products from industry or municipal wastewater solids (Hackett et al., 1999; Villaseñor et al., 2011). Traditional methods of organic waste management and disposal, such as landfilling (Louis, 2004) and incineration (Assamoi and Lawryshyn, 2012; Cherubini et al., 2009) are unsustainable and a transition toward prevention, recovery, diversion, and valorization of organic matter, as outlined by the food waste/recovery hierarchy (US EPA, 2015a) has occurred. For instance,

modern landfill designs not only incorporate leachate collection and treatment, but also methane capture as a bioenergy source (Adhikari et al., 2006; Behera et al., 2010; Themelis and Ulloa, 2007). Many waste management authorities around the world are moving to ban landfilling of organic wastes (Behera et al., 2010; BioCycle, 2018; California Legislature, 2014; Government of Nova Scotia, 1994; Government of Ontario, 2017). In Canada, some jurisdictions have either banned the landfilling of organic waste (Nova Scotia and Prince Edward Island) or are in transitional phases (Ontario) to do so (Government of Ontario, 2017). Despite this progress, food wastes (edible and non-edible) contribute 23%, or 6 million tonnes, of the total Canadian municipal solid waste (MSW) stream sent to landfill (Environment and Climate Change Canada, 2020). Organics recycling programs are now aimed at diversion and conversion into new products or developing energy

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recapture systems, i.e., anaerobic digestion, in order to reduce environmental impacts, on water resources and the atmosphere, while also extending the lifespan of landfills (Kulikowska and Klimiuk, 2008). Furthermore, municipal and commercial sectors are valorizing organic wastes through the processing or conversion of organic material into marketable products, e.g. compost, growing media (Kabongo, 2013). Organic waste generation is increasing at unprecedented rates due to rapid urbanization and economic development globally (Muller, 2009). Globally, from 2007 to 2017, the top five main crops (sugar cane, maize, wheat, rice and potatoes) have seen an increase in production both on account of food demand and for biofuel production, especially with maize (FAO, 2019). Maize production increased from 792 million tonnes in 2007 to 1.13 billion tonnes in 2017, while the other main commodities also experienced sizable increases: 606 to 771 million tonnes (wheat), 656 to 769 million tonnes (rice) and 314 to 388 million tonnes (potatoes). Similar production trends can be seen in meat, dairy, and egg products, with chicken numbers increasing by 4.5 billion head in 2007 to 22.8 billion in 2017. Similarly an increase in fish production has gone from 98 million tonnes in 1990 to 173 million tonnes in 2017, predominantly driven by the expansion of aquaculture (FAO, 2019). Global food loss and food waste (FLW) ranging from unavoidable food production losses to avoidable waste, (including discards and unpurchased material) is estimated to be 1.3 billion tonnes according to 2019 FAO statistics, and within North America, the USA generates an estimated 168 million tonnes of FLW (FAO, 2019), while Canadian FLW is estimated to be 34.9 million tonnes (Gooch et al., 2019).

In agriculture, animal manures and slurries have an intrinsic value to the agricultural community, providing a source of energy and nutrients in the form of nitrogen, phosphorus, potassium, and micro-nutrients for crop production. Similarly, food wastes have value past their original purpose of human (or animal) consumption, as a source of energy and nutrients in a range of valorization technologies. Anaerobic digestion (AD) is one favoured approach for treating food wastes, using biogas as a waste-to-energy fuel, with the resultant digestate land-applied for crop production. The European anaerobic digestion market is projected to be worth CAD\$94.2 billion by 2026 (Global Market Insights Inc, 2020), demonstrating the tangible economic value of recycling organic waste materials. Anaerobic digestion can be a valuable and important method of utilizing waste material, already an important component of the European energy mix (International Renewable Energy Agency, 2018). A significant disadvantage of AD technologies is the large capital-intensive infrastructure and government-subsidized electrical supply rates, as well as a local land-base for disposal of the digestate. In North America, the cost of infrastructure and jurisdictional variability on subsidizing or supporting electrical generation from alternative energy sources are still barriers for wide-scale adoption of AD (Canadian Biogas Association, 2019). Other routes for organic food wastes include rendering of animal by-products (for protein meal and oil products), autoclaving and hydrolysis, incineration (or waste-to-energy), pyrolysis, gasification, and plasma gasification. Each of these methodologies have benefits and can produce valuable products, but are also associated with constraints on the raw material composition, are capital-intensive to build and operate, and at best are more suited to large-scale industrial settings where volumes of waste material generated are high, land is expensive, and energy recovery is viable (NI Business Info, 2012).

Composting is a viable process to optimally manage the biological decomposition of organic matter under aerobic conditions. As an aerobic decomposition process, the resulting predominant products include stabilized organic matter (compost), water, heat, water vapour, and carbon dioxide (Diaz and de Bertoldi, 2007; Epstein, 2017). The quantity of CO₂, and of other gases (N₂O and CH₄), produced will be influenced by a number of factors including aeration rate, mixing rate, feedstock composition, C:N ratio (Jiang et al., 2011), moisture content (El Kader et al., 2007), and the temperature profile during the process. Aerobic composting is achieved through a number of different processes, ranging from outdoor windrows or static piles to indoor in-vessel systems (US

EPA, 2015b). Han et al. (2018) evaluated the influence of ambient temperatures on greenhouse gas generation from sewage sludge processed in a large-scale municipal composting facility. Evaluated over three seasons (winter, spring and summer), Han et al. found clear variation in CO₂ emissions between the three different seasons, finding higher total CO₂ emissions in spring followed by summer, and winter, 941.72 mg CO₂ kg⁻¹ h⁻¹, 698.97 mg CO₂ kg⁻¹ h⁻¹ and 220.54 mg CO₂ kg⁻¹ h⁻¹, respectively. Compost has an established commercial and agricultural value as a soil amendment, nursery growth media, and as an approach to deal with organic wastes across a range of management scales (Hubbe et al., 2010; Meyer-Kohlstock et al., 2013; Walker et al., 2006). In the USA, composting comprises approximately 8.5% of MSW (292.4 million tonnes) management processes (US EPA, 2017) and is considered a core process, along with other technologies such as anaerobic digestion and incineration (Ayilara et al., 2020; US EPA, 2017; Razza et al., 2018). Municipal composting is challenging due to the heterogeneity of material, high requirement for labour, time, and cost to separate contaminants such as plastics, glass, and other materials, but also due to the difficulty in controlling the source material. The final product varies in quality and consistency between jurisdictions leading to a price per tonne of compost ranging from CAD\$30 to \$130. Increasingly, the economics of municipal organics diversion programs face significant challenges ranging from high labour, infrastructure, and transportation costs, to low market value for composted products. Pandeyaswargo and Dickella (2014) evaluated the economics of municipal composting in Asian cities, determining that factors such as: location, scale, processing methods, technology used, end-product quality, tipping fees, government incentives or rebates, and the ability to find a suitable market for the end-product drive financial feasibility. Coker (2010) found that controlling costs and maximising processing capacities is fundamental in ensuring profitability of municipal composting facilities (Coker, 2010), while the NorthEast Recycling Council stated that 87% of US composting operations fail within five years without government subsidies (Castagnero, 2014). Development of circular economies, with emphasis on improved valorization of bio-waste streams, can generate new opportunities to reduce operational costs, improve efficiency of utilization, and build synergies with different sectors of the economy (Masullo, 2017).

Composting of organics is a vital step in reducing greenhouse gas emissions, reducing the impact of organics going to landfill and pushing forward the concept of a circular economy where little to no material is deemed valueless (Favoino and Hogg, 2008; Lou and Nair, 2009). The economics of commercial composting systems comprises: 1) the initial capital investment for infrastructure and machinery costs, and, 2) ongoing expenses including labour, maintenance, utilities and local and provincial waste disposal rates. Revenues are generated through gate (tipping) fees and the sale of compost. Composting has become not only more profitable due to the use of gate fees, but due to higher gate fees associated with landfilling. The 2018 median gate fee for landfilling within the UK was found to be CAD\$182 tonne⁻¹ compared to CAD\$78 tonne⁻¹ for in-vessel composting and CAD\$40 for windrow composting (WRAP, 2019). Murphy and Power (2006) evaluated four municipal composting systems in Ireland, finding that the greater the size of the system, the more economically viable they became. Profits generated were shown to be predominantly from gate fees, although the sale of compost also provided moderate contributions to revenues from these systems (Murphy and Power, 2006). Similarly, Pandeyaswargo and Premakumara (2014) evaluated the financial sustainability of six composting facilities in developing Asian countries, finding that medium and medium-large facilities were the most viable, concluding that gate fees and the sale of compost were important in making the facilities economically viable; however, the selection of composting technology, management practices, market access, and availability of carbon credits were also key. Organic matter, from pre- and post-consumer food wastes, animal and agricultural wastes, and municipal wastewater solids have intrinsic value as sources of energy (carbon) and nutrients (Pham

et al., 2015; Uçkun Kiran et al., 2014; Zhang et al., 2014). As a feedstock for the composting process, organic matter provides both carbon, as an energy source and major building block for microbial metabolism and growth, and nitrogen for production of protein, nucleic and amino acids, and enzymes for cell growth and function. Typically, a C:N ratio of 25–30:1 for initial feedstock mixtures for composting is considered ideal for supporting these microbial processes.

Rapid expansion of controlled environment agriculture (CEA) in recent years provides a new pathway toward food security and resource recovery (Hagbert and Bradley, 2017). Innovative practices for year-round CEA, as exemplified in the proliferation of vertical farming systems, has clear potential to valorize and use residual organic biomass from these systems toward improving production efficiencies and reducing operational costs. Vertical farming was valued at CAD\$2.89 billion in 2018 and is estimated to reach CAD\$16.05 billion by 2026 (Patil and Baul, 2019). Controlled environment agriculture systems, such as vertical farms, are energy and capital intensive, integrating multiple components such as, HVAC for environmental control of temperature, humidity, and ambient gas composition, and electric lighting systems. Plant growth is modulated through the use of targeted spectral compositions using energy efficient LED lights and in many facilities through carbon dioxide gas enrichment of the growing environment to increase baseline productivity. These systems provide the benefit of year-round production, including some reduction of the carbon footprint associated with long distance transportation, but comes at the expense of higher energy usage and cost. In contrast, large traditional greenhouse operations, including some with hydroponic systems, are increasingly using technologies found in vertical farming systems to improve production, including LED lighting systems (supplemental), solar capture, water capture and reuse, and general improvements in controlled environment systems (Kalantari et al., 2017). Integrating technological advances in recapture and utilization of CO₂ from composting of organic wastes can minimize the carbon footprint of CEA operations and improve plant productivity rates. The vertical farming sector (both large and small operations) is estimated to reach CAD\$5.91 billion globally in 2021, and projected to reach CAD\$26.12 billion by 2028 (Grand View Research, 2021). Vertical Farm systems are touted as being more water efficient (Avgoustaki and Xydis, 2020) and productive than conventional agriculture, while also reducing the carbon footprint associated with food transportation (Al-Kodmany, 2018; de Oliveira et al., 2020). Inherent advantages of vertical growing systems, and more broadly CEA in general, are the capacity to recycle nutrients more efficiently, improved water use efficiency through capture and reuse of irrigation/fertigation solutions, and reduced the use of pesticides. The advantage of locally grown foods includes a reduced transportation carbon footprint, and fresher products that are closer to market, and thereby typically have longer [consumer] shelf life. Further, there is also more direct support of local economies when production is based within a given community. Global production of lettuce ranges from 26 to 29 million tonnes over an area of 1,227,358 ha, with an estimated field-based production between 2.2 and 3.2 kg lettuce m⁻² yr⁻¹ (Naus, 2018; Shatilov et al., 2019). In contrast, VF systems are estimated to generate 41 kg lettuce m⁻² yr⁻¹, which is 13–18.5 times more production per square metre (Barbosa et al., 2015; Benke and Tomkins, 2017). However, if lettuce root biomass is approximately 10–20% of total fresh biomass under aeroponic or hydroponic production, which would equate to 4.1–8.2 kg root biomass m⁻² yr⁻¹, waste biomass generated in VF systems under current annual yields would be 2.9 to 5.8 million tonnes per year (Li et al., 2018). Under conventional agriculture, the root biomass would remain in the soil, recycling the carbon back into the system. However, in CEA or VF, current industry practices for residual biomass include disposal at local waste facilities or landfills and offsite composting where available. The cost of landfilling or recycling will vary based on local gate fees, accessibility to centralized organics composting facilities, cropping cycle, local by-laws, and the size of the vertical farm. Disposal costs for CEA or VF facilities will depend also on

crop type, the amount of non-commercial residual biomass generated, (i.e., root mass, stalk/stems/leaves) and the ratio of harvestable to non-edible biomass (harvest index). Another significant implicit challenge in these production systems is the use of fossil fuels or carbon-negative sources to enrich the growing environments with CO₂ (Esmeijer and Bloem, 1999; Ghiat et al., 2021). The overall cost of CO₂ is not necessarily a major factor (in the absence of a carbon tax) but is generated through consumption of fossil fuels (Ghiat et al., 2021; Poudel and Dunn, 2017). Atmospheric enrichment of the production facility beyond ambient CO₂ can dramatically increase production and improve profitability.

On-site processing of waste biomass in CEA or VF systems using in-vessel composting may have value by reducing third-party collection, transport, and gate fee costs. Moreover, the potential exists for integration of in-vessel composting to utilize metabolic carbon dioxide in plant growth rooms and to generate final compost media for use as a nutrient source or seed growth media. With 592 cultivation licences granted in Canada, the recent rapid growth of cannabis production facilities across Canada opens new opportunities for the recycling of waste biomass into gas and heat by-products that can be utilized within the closed growing environments (Health Canada, 2018). Many CEA cannabis operations incorporate CO₂ enrichment of the production chambers using non-renewable resources. The capacity to integrate these two types of technologies currently exists but economies of scale related to soil-based production agriculture have been an economic bottleneck. Currently, shifting public perception of the environmental impacts from soil-based agriculture, increased demand for high value fresh produce throughout the year, and the vulnerabilities in global food supply chains exposed by the Covid-19 pandemic are opening up the opportunities to capitalize on this technology integration (Market Line, 2020) (Storey and Heliospectra, 2021).

The scientific literature on composting focuses primarily on chemical dynamics of open-air/open access composting systems and/or on the generation of compost products (Reyes-Torres et al., 2018). A significant gap exists within this body of scientific literature on how the composting process and available technologies can be adapted to recapture CO₂ gas for utilization in other systems, such as vertical farms or CEA. This review aims to highlight the potential to valorize by-product gas, specifically carbon dioxide (CO₂) from the composting of organic wastes from municipal, commercial, or agricultural sources for use in vertical farming and other controlled environment agriculture production systems.

2. Methodology

To achieve a comprehensive review and analysis of data on CO₂ emissions from composting, a systematic evaluation of the scientific literature and government publications was undertaken (Fig. 1). Scientific databases used to identify suitable literature for inclusion in this study were accessed and included: Google Scholar (<http://scholar.google.ca>), Science Direct (<http://sciencedirect.com>), SpringerLink (<http://springerlink.com>) and Web of Science (<http://webofknowledge.com>).

Utilizing database search functions, key search terms were input based upon defined parameters: 1) the main metric relating to CO₂, 2) the process of CO₂ generation, 3) materials used in compost production, and 4) the composting systems being evaluated. The search parameters were broken in to three stages, with incremental specificity at each stage (Fig. 1).

Each database provided a numerical value of results based on the search parameters. Based upon the number of results, additional filtering was applied to narrow the search results and remove articles with materials not deemed to be as relevant. For example, an overlap exists in the application of composts to soils with the use of biochar; however, since the focus is on process generated CO₂ rather than carbon sequestration, journal articles where biochar was part of the composting

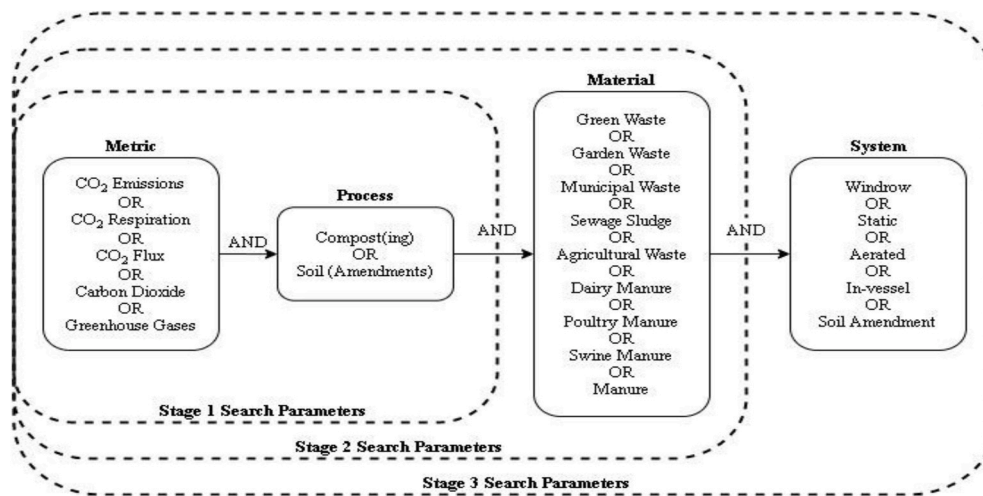


Fig. 1. Literature review three stage search parameters and keywords.

material or used as a soil amendment were removed (Fig. 2).

2.1. Data reporting

A challenge in reviewing and synthesizing the literature on CO₂ emissions from composting is the large variation in how data are measured, and units presented across studies. A number of different analytical techniques and instrumentation are available for the capture and measurement of CO₂ gas, including gas chromatography, portable gas analysers (IRGA), NSDAIR sensors, and alkali traps and back-titration using acids. While the various approaches can provide comparable results, there may be variations in the sensitivities and in the format that data are presented. The data evaluated in this review originate from studies conducted under open and isolated conditions, which affect the extrapolation of gas production in real-world settings. This review classifies open systems as any system that is open to the atmosphere such as, windrows, aerated piles, and static piles. Closed or isolated systems include material enclosed within a vessel either at a commercial, pilot, or lab/bench-top scale.

A commonality in reporting units of CO₂ is important for comparing different data sets. For instance, many studies expressed respiration in terms of cumulative emissions (g CO₂ kg), daily (g CO₂ kg⁻¹ d⁻¹), or

hourly, (g CO₂ kg⁻¹ h⁻¹) emissions. Where data was not presented in these units, values were converted into a standard format provided other data were available in the manuscript, specifically the weight of the material being composted (wet weight or dry weight) and the duration of each study. This review presents other units, grouped together, when conversion was not possible. In some studies, data has been reported on an area or flux basis, i.e. g CO₂ m⁻² d⁻¹ or g CO₂ m, and were used when the results could be directly compared to other research. In some cases, gas units were presented without supporting data to allow for conversion and comparison, g CO₂ or g CO₂ d⁻¹, but were used when general trends could be inferred. This review further sought information relating to peak CO₂ emissions; data representation is based upon the point in the composting process where the maximum CO₂ emissions occurred and linked to whether the system was open or closed. Differentiation has been made between peak CO₂ emissions from open and closed systems.

3. Overview of the decomposition process and composting methodologies

The aerobic decomposition of organic matter occurs as biochemical transformations of carbon and other elemental components, mediated through an array of bacteria, fungi, and actinomycetes (Trautmann and

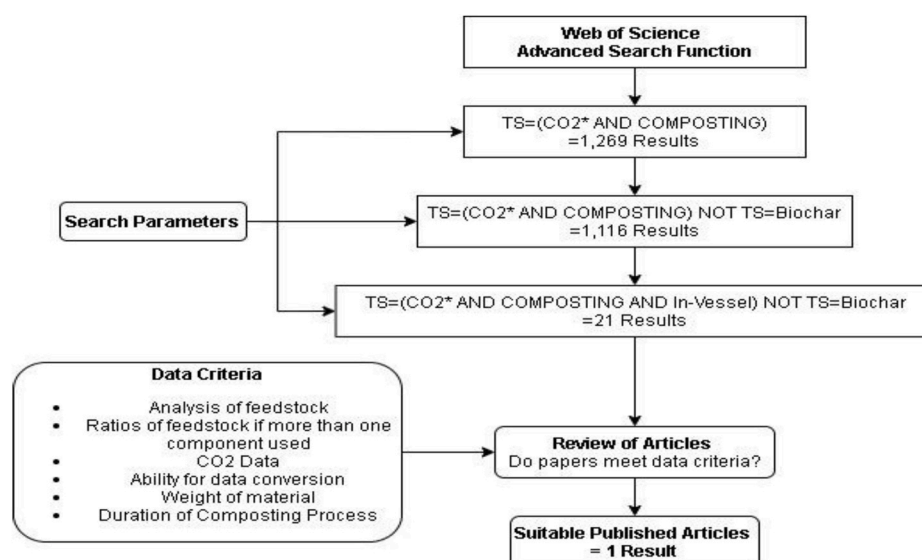


Fig. 2. Filtering steps and process flow for research literature search, e.g. Web of Science.

Olynciw, 1996). Compost systems contain a large diversity and number of microbial organisms, with bacterial species typically dominating, followed by fungi, and actinomycetes. Dominant phyla in compost systems typically consist of Actinomycetes, Bacteroidetes, Proteobacteria, Firmicutes, and Sordariomycetes (Huhe et al., 2017). Microbial groups will degrade compounds based on energetic paths of least resistance, typically in the order of carbohydrates, fats, proteins, cellulose and hemicellulose, and lignin. Oxidative breakdown of primary compounds to secondary metabolites and constituent chemicals are catalyzed by enzymes released onto surfaces or in the pore solution (Huhe et al., 2017). A proportion of the organic carbon is released as carbon dioxide during the process of oxidative decomposition, typically 50–60% of the total carbon is lost over time (Chang et al., 2019). The rate and quantity of carbon loss during organic matter degradation in compost systems varies as a function of management practices (Tiquia et al., 2002) employed to ensure adequate levels of available oxygen, unsaturated moisture conditions, and balancing the C:N ratio of the mixture (USDA - Natural Resources Conservation Service, 2007). The consumption of O₂ and the production of CO₂ is a clear indicator of the aerobic processes working within the compost system, while CH₄ is an indicator of anaerobic processes and N₂O and indication of nitrification (USDA - Natural Resources Conservation Service, 2000).

Composting can be divided into three phases of microbial activity: 1) active, during which labile carbon is rapidly mineralized into CO₂ or immobilized back to organic forms, 2) stabilisation, during which the remaining carbon compounds are either more recalcitrant to degradation or more complex forms of carbon are created that are also resistant to decay, and 3) maturing/curing, during which the final product of the process yields a material that consists of organic and inorganic forms of carbon that are in pseudo-equilibrium leading to a low rate of carbon mineralization. The time frames for these phases of composting can range from several months to years depending on the type of system, the intensity of management, aeration conditions, and environmental conditions (Bidlingmaier et al., 2004) (Table 1).

3.1. Composting system design overview

There are different methods used for composting, from large-scale windrow systems that are open to the environment, commonly found in municipal systems, to closed in-vessel systems that having some type of housing or container offered in a variety of sizes and requiring a reduced footprint. There are four main types of composting systems that are conventionally used, these being windrows, aerated static piles, enclosed channel, and in-vessel systems, with a fifth less common system, hyperthermophilic composting (Campbell and Glasser, 2009; Grunklee, 1998). Management objectives for processing organic wastes through composting include: conversion of biologically unstable organic wastes into more stable forms, reduction of the volume of organic waste, improving sanitation conditions through the elimination of pathogens or hazardous chemicals (hyperthermophilic), and retention of nutrients in organic forms for reuse in agricultural or other production systems (Groat, 2017; Rihani et al., 2010). Composting systems, regardless of their scale, aim to provide an environment where aerobic decomposition

is optimized to produce a stable end-product. Composting systems increase in technical complexity from windrows to in-vessel systems because of the level of external controls imposed on the process, such as active aeration, automated control of mixing, enclosures that monitor and manage temperature, and leachate collection.

3.1.1. Open composting systems

Open composting systems are open to the atmosphere and subject to the effects of the local climate, seasons and weather, impacting the rate of decomposition and time to achieve a stabilized homogenous compost product but can be conducted at a large scale (Composting Council of Canada, 2006).

3.1.2. Windrow composting systems

Windrow systems are systems open to the environment and are mechanically mixed using tractor drawn windrow turners or front-end loading buckets. The mechanical mixing is conducted to enhance aeration, increase exposure of labile components located on the outside of the pile, and to manage temperatures. The length of compost windrows is typically determined by site and management factors, providing significant flexibility to compost facilities, but width and height are generally each restricted to 3–4 m for effective composting and easy access by normal farm equipment (CH2M HILL, 2013; Composting Council of Canada, 2006; Groat, 2017). Windrow composting is heavily dependent on operator skill, with operators monitoring the pile and turning as necessary, with more frequent turnings leading to a faster composting process. These systems are suitable for facilities receiving very high volumes of heterogeneous feedstocks on a regular basis, i.e. large municipal facilities, and where a large land footprint is available.

3.1.3. Aerated static piles

Aerated static pile composting is comprised of single batch-based trapezoidal piles of heterogeneous feedstocks. Depending on the feedstock type, the materials may be layered, such as in animal mortalities, or mixed prior to establishing the structure. These compost systems may be aerated passively or by forcing (positive) or pulling (negative) air through the trapezoidal compost pile. Once established, agitation only occurs when piles are combined or moved to a different area for curing. Piles often are covered with a layer of finished compost or wood chips, which are then incorporated when the piles are moved. Static aerated piles are generally used for biosolids, animal mortalities, or feedstocks that have similar consistency. Enclosed aerated static piles operate under the same principles as aerated static piles but are enclosed predominantly by plastic permeable membranes or organic matter is bagged and forced air is applied (CH2M HILL, 2013; Composting Council of Canada, 2006; Groat, 2017). Aerated systems are typically found in the agricultural sector for management of crop residues or animal mortalities. They are established as a low maintenance and relatively low-cost management system.

3.1.4. Closed composting systems

Closed composting systems are those not open to the atmosphere and are either covered or contained or enclosed within a structure or vessel. These systems often incorporate mechanical components for mixing and managing the decomposition process resulting in a shorter time-frame for decomposition (Composting Council of Canada, 2006).

3.1.5. Enclosed composting systems

Older in-vessel composting systems, referred to as in-vessel bays, enclosed halls, or silos, are typically an enclosed or semi-enclosed building with mechanical components that move material from one end of the structure to the other. Enclosed channel systems operate using mechanical agitation, with organic matter placed in long channels with concrete walls. Turning machines are utilized to agitate the material and move it from one end of the system to another, where the decomposition process takes place throughout. Forced aeration is typically through

Table 1

Typical durations for each composting processing phase in three different techniques (Composting Council of Canada, 2006).

Process Type	Windrow	Aerated Static Pile	In-vessel
Processing Phase	Duration (days)		
Active	16–40	16–30	4–15
Stabilisation	30–60	30–60	21–45
Curing	Up to 8 months (turned)	1–3 months (aerated)	1–2 Months (turned)
Total Time	2–12 months	2–6 months	1–4 months

vent holes in the floor agitating the bed (CH2M HILL, 2013; Composting Council of Canada, 2006; Groat, 2017).

3.1.6. Vertical in-vessel silo systems

Vertical in-vessel silo systems offer a passive aeration system for cage-suspended organic matter within a silo vessel. These systems have several challenges related to odour control, build up of leachate in air-flow channels, and the slow rate of decomposition due to lack of agitation. Modified systems include the incorporation of aeration tubes within the piles to aid in passive aerations. Critical to these systems is the homogeneity of the feedstock with regards to particle size to ensure an even composting process, therefore rotary drum systems may be utilized as a pre-treatment.

3.1.7. Modular in-vessel systems

Modular in-vessel composting systems are fully enclosed and automated systems requiring direct energy input and are designed to provide optimal composting conditions and feedstock homogenization, resulting in an accelerated degradation process. The resulting output depends on the system type, material retention time, feedstock types, and operating conditions. The product may potentially require additional curing for a final, stable product. In-vessel composting systems are more expensive based on initial equipment requirements but require a smaller land base and range in processing capacity (CH2M HILL, 2013; Composting Council of Canada, 2006; Groat, 2017; Tetra Tech EBA inc., 2014). Modular in-vessel composting may include environmental monitoring, gas and temperature sensors, and mechanical mixing of material, along with active aeration to provide more control over the composting process (Fig. 3).

Small modular in-vessel systems are designed to process 10 tonnes per annum, medium systems can typically process up to 100 tonnes per annum and large systems can process approximately 5,000 tonnes per annum. Processing capacity for many of these modular systems can be scaled up through addition of modules. Scaling up from small systems to large systems results in increasing complexity with respect to system design, operational input, running costs (e.g. electricity consumption), and logistics (Tetra Tech EBA inc., 2014).

These systems can take up a significantly lower footprint than open composting systems but require higher initial capital investments. Different systems may have a fixed capacity with limited scalability, while other provide some flexibility on control of processing capacity to allow for increased handling and processing capacity through the installation of additional modules. A number of in-vessel designs and flow through approaches are currently on the market ranging from non-mechanically agitated (static) in-vessel containerized composting systems to fully automated turned and air injected systems at a range of different scales and configurations allowing for stand-alone or multi-unit integration (Engineered Compost Systems, 2019; Green Mountain Technologies, n.d.; Brome Compost Inc., 2019; Renewable Carbon Management LLC, 2019; Global Composting Solutions Ltd., n.d.). Different composting approaches have advantages and disadvantages that come with the operational goals and anticipated valorization pathways (Table 2).

Table 2

Advantages and disadvantages of open and closed composting systems. (Adapted from (Bidlingmaier et al., 2004; Nichols, 2009).

Open System		Closed System
Windrow	Aerated Static Pile – Positive Pressure	In-vessel
Advantages		
• Low cost	• Effective heat removal	• High rate of composting
• Simple operation	• Low land requirements	• Off-gas attainment
• Acceptable compost quality	• Faster decomposition	• High degree of control
	• Reduced incidences of odour	• System monitoring
		• Fast processing of harmful material
Disadvantages		
• Low level process control	• High capacity blowers required	• High investment and operating costs
• High land requirement	• Potential stratification of material	• Skilled operators
• Dust generation	• Potential leachate	• Potential leachate
• No ability to manage off-gasses	• Off gases to be managed (if indoors)	• Off-gases to be managed

4. Cycling of carbon from organic wastes during composting

Conventional practice in composting aims to generate a biologically stable compost end product. Managing the entire process for extraction of valuable products, such as gas and heat, is a fundamental shift in the organic waste management paradigm. This paradigm shifts includes capture and recycling of process by-products from composting such as energy and CO₂. Transforming the model of composting operations toward a circularized management approach requires: 1) identification of sinks for heat and CO₂, 2) quantifying heat and CO₂ generation per mass unit of organic matter, 3) design and integration of capture and distribution systems for heat and gas, and 4) optimization of composting technologies for whole system recapture of resources. This review will focus on the generation and recycling of CO₂ gas from composting as a valorization pathway.

4.1. Factors influencing CO₂ production in composts

CO₂ emissions are a recognised metabolic by-product of the composting process, along with other gases including, NO_x, NH₄ and CH₄. Gas type or composition is generally managed through effective composting, maintaining aerobic conditions, anaerobic by-products and feedstock selection. In most circumstances, the goal in composting is to generate a stabilized organic matter product and therefore, little research is conducted on the quantification and capture of CO₂ for use. Several variables influence the rate and quantity of CO₂ produced over the active composting phase. The main factors ensuring successful composting and CO₂ production are C:N ratio, moisture content, oxygen availability, temperature, and pre-processing (i.e., grinding, homogenizing, particle size). Other factors impacting CO₂ generation include mixing rate, bulking agents, compost amendments, and feedstock composition.

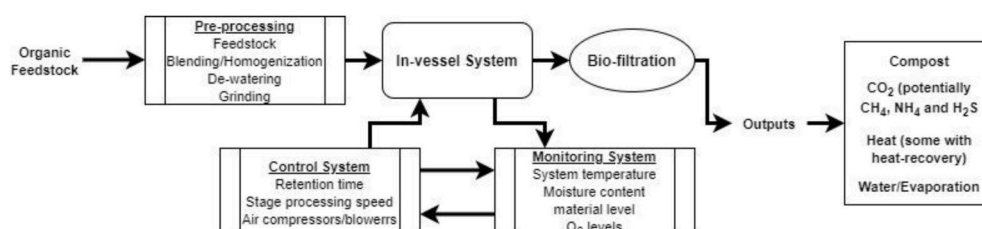


Fig. 3. Generic system flow model for a modular in-vessel composting system with inputs and outputs.

4.1.1. Feedstock composition

Most of the composting studies identified through this review focus on agricultural manures or sewage sludge as a basis for nitrogen amendments, with a smaller number of studies focused on green waste (Table 3). The majority of carbon feedstocks in compost originate from wood shavings or sawdust, straw, rice husks, and hay. It is understandable that agricultural waste and sewage sludge predominate in the literature given the scale of the agricultural sector, followed by municipal green waste. Maximising CO₂ emissions, or identifying the potential for CO₂ emissions from specific sectors where high volumes of waste can be found (e.g., cannabis production, vertical farms or CEA) is an area requiring further research. Feedstock composition, both physical and chemical, drives chemical reaction rates and pre-processing requirements (i.e. grinding, chopping) through availability of labile carbon and nitrogen. Physical properties of feedstocks will influence surface area reactions, retention of water, and porosity.

4.1.2. Influence of amendments on CO₂ production

A number of studies have evaluated the influence on gas emissions from compost systems using supplements or microbial inoculants to common compostable material (Chen et al., 2015; Mao et al., 2019; Peng et al., 2019; Ren et al., 2019). The primary focus of these studies has been to evaluate the influence of supplements or amendments on microbial activity, to improve the fertilizing quality of the finished compost (generally total nitrogen content), to reduce GHG emissions, or a combination of the three. Studies have ranged across both open (windrow or static pile systems) and closed (in-vessel, or lab-scale bioreactors) systems.

Bean dregs, a by-product of tofu production, are classified as a nitrogen rich material with large particle sizes and poor storage properties, making it a useful material for compost addition. Yang et al. (2020)

Table 3

Literature values for cumulative CO₂ emissions from selected composting studies using different feedstocks.

Material	No. of Days	g CO ₂ kg ⁻¹	Study Type	Reference
Straw bedding and beef manure	99	165	General Analysis	Hao et al. (2004)
Wood-chip and beef manure	99	145.6		
Dairy manure	42	587	Turning Analysis	Mulbry and Ahn (2014)
Dairy manure	42	686		
Dairy manure	42	699		
Dairy manure	42	627		
Dairy manure	42	722		
Animal manure	28	62.3	Bulking Agent Analysis	(M.A Chowdhury et al., 2014)
Animal manure and plastic tube	28	63.2		
Animal manure and woodchips	28	48		
Animal manure and biochar	28	41.2		
Animal manure and barley straw	28	50.3		
Animal manure and lupin residue	28	61.2		
Cattle Slurry	31	93	Aeration Analysis	(M.A. Chowdhury et al., 2014)
Cattle Slurry	31	85		
Poultry manure	31	79		
Poultry manure	31	70		
Sewage sludge	60	91.23	Temperature and Aeration Analysis	Han et al. (2018)
Sewage sludge	60	179.9		
Sewage sludge	60	216.26		
Pig manure	28	358.27	Amendment Analysis	Zhang et al. (2017)
Pig manure	28	234.9		

evaluated the impact of bean dregs as an amendment to pig manure composting with the purpose of evaluating the carbon and nitrogen losses through GHG emission. Using wheat straw and pig manure as the carbon and nitrogen sources, bean dregs were added to the compost material using lab-scale 130L bioreactors (in-vessel system) and composted for 56 days, with weekly turning. CO₂ emissions peaked within the first 10 days of the composting process, 393.3 g CO₂ d⁻¹ for the amended compost mixture, and 352.7–393.3 g CO₂ d⁻¹ for the control. The cumulative CO₂ emissions over 56 days were 4291 g CO₂ for the bean dreg supplemented compost and 3432 g CO₂ for the control (Yang et al., 2020). In another study, Mao et al. (2019) evaluated the impact of several supplements to pig manure compost, with the aim of evaluating changes in microbial communities within the compost and the impact on greenhouse gas emissions (Mao et al., 2019). Using sawdust and pig manure as the carbon and nitrogen sources, they evaluated the impact of zeolite, medical stone (a mineral additive), and wood vinegar. The study used small-scale glass bottle reactors and ran for 36 days with CO₂ emissions peaking on day 9, at 37.9 g CO₂ d⁻¹ for medical stone, 32.8 g CO₂ d⁻¹ for zeolite and 40.9 g CO₂ d⁻¹ for wood vinegar. Peng et al. (2019) also evaluated the impact of zeolite, as well as superphosphate, combined with ferrous sulfate on GHG emission reductions and improvements to fertilizer efficiency from composted chicken manure with rice husks under windrow composting. Using a standard windrow composting system, turned every two days for the first 15 days followed by 31 days of no turning, they identified peak CO₂ emissions during the first 15 day phase, with an increase in oxygen diffusion and the promotion of aerobic decomposition caused by the turning process. The addition of superphosphate reduced cumulative CO₂ emissions by 27.6% compared to the control. Another study evaluating addition of impact of superphosphate with pig manure also reported reductions in O₂ consumption and CO₂ production, with a mean production rate of 14.05 g CO₂ kg⁻¹ DM d⁻¹ and 20.6 g CO₂ kg⁻¹ DM d⁻¹ for amended superphosphate and the control, respectively (Zhang et al. 2017). On a cumulative basis, the control treatment, using no amendments, from Zhang et al. (2017) produced 358.27 g CO₂ kg⁻¹ compared to the superphosphate amended compost generating only 234.94 g CO₂ kg⁻¹. Chung et al. (2018) evaluated the influence of another clay additive to a wheat straw and poultry manure compost on GHG emissions, reporting reduced N₂O and CH₄ with increasing ratios of clay additive (DM basis) to compost but CO₂ emissions also increased. These studies were conducted using bench-top aerobic bioreactors over a 50-day period with mixing occurring after the 3rd day then every 7th day. Other studies have evaluated diatomite as a means to reduce GHG emissions (Ren et al. 2019). Using sawdust and poultry manure, their study added 5% and 10% diatomite (DM basis) to the compost mixture and composted the materials in a closed 80L container. Cumulative CO₂ emissions were found to be 1421.04 g, 1587.06 g, and 1840.44 g for the control, for 5% diatomite, and 10% diatomite, respectively. The addition of diatomite increased CO₂ production by 11.7% and 29.5%, and resulted in reductions of gases with higher global warming potential than CO₂, namely NH₃, N₂O, and CH₄ (Ren et al., 2019).

The addition of different reactive substrates can impact microbial processes occurring within the composting system, influencing the quantity and nature of gases produced. Reflecting on capturing CO₂ emissions from closed aerobic bioreactors, the use of select amendments may be a valuable method of influencing the rate and yield of CO₂ gas, while simultaneously reducing other GHG emissions. Based on a review of the literature, there still is a clear need to evaluate the use of amendments in more detail, identify regionally accessible amendments, and conduct studies at a larger scale. Increasing CO₂ levels while reducing other greenhouse gases has a two-fold benefit: 1) higher quantities of CO₂ for capture and utilization, and 2) reduce the burden on gas filtration systems, thereby lowering operational costs.

4.1.3. Effect of aeration and movement of gases through compost systems

There have been a small number of studies focused on the

distribution of gases being emitted from windrow systems. Andersen et al. (2010), aimed at quantifying greenhouse gases emitted from standard windrow systems and evaluated the emission patterns of CO₂, CH₄ and N₂O (Andersen et al., 2010). Based on a 9 m wide, 4 m high and 115 m long windrow system, operating for 409 days, the researchers measured cross sectional readings for different GHG being produced. They found CO₂ was predominantly produced in the central portions of the windrow (26–32%) and lower at each side of the windrow (12–14%) with corresponding lower levels of O₂ (4–6% central, 10–12% sides). An increased level in gases in the central column of the windrow, aligns with the general chimney effect occurring within a windrow system, with air being drawn in from the sides and funneling up through the central column (Andersen et al., 2010; Avnimelech et al., 2013). Ma et al. (2020), evaluated the effects of intermittent aeration on greenhouse gas emissions and bacterial community succession in a large-scale covered compost and reported a reduction in emissions with increased aeration rates. Several studies corroborate these results, supporting that effective mixing and aeration ensures full aerobic degradation occurs evenly across the compost system (Ahn et al., 2011; Chowdhury et al., 2014; El Kader et al., 2007; Jiang et al., 2011). One study examined different aeration methods and aeration rates on greenhouse gas emissions using corn stalks and pig manure at a pilot-scale level (Jiang et al. 2011). Aeration rates of 0, 0.18, 0.36 and 0.54 L min⁻¹ kg⁻¹ (DM) and using a continuous and intermittent mixing regime (turning every two weeks) over a 10-week period. The study found that increased aeration rates resulted in higher NH₃ and N₂O emissions but reduced CH₄ (Jiang et al., 2011). The purpose of mixing or turning compost is to increase the availability of O₂ within the system thereby reducing anaerobic activity. Mixing will further alter the temperature of the compost pile, causing a drop in temperature throughout the pile. Mulbry and Ahn (2014) evaluated different turning regimes over a 42-day trial period, including turnings on days 14 and 28 (686 g CO₂-C kg⁻¹ VS degraded), on days 21 and 35 (699 g CO₂-C kg⁻¹ VS degraded), on day 28 (627 g CO₂-C kg⁻¹ VS degraded), on day 35 (722 g CO₂-C kg⁻¹ VS degraded) and a static pile (587 g CO₂-C kg⁻¹ VS degraded).

4.1.4. Novel approaches impacting CO₂ emissions

There have been a limited number of novel approaches that have been applied to composting systems, with the aim to improving compost maturity and reducing GHG emissions. Tang et al. (2019) developed a process of electric-field-assisted aerobic composting. In this process, a direct-current was applied to the compost system with the purpose of using the electric current to increase O₂ uptake. The theory relates to the production of electrons during microbial activity (respiration), with O₂ being the terminal acceptor. Oxygen is a vital component of the composting process, however Tang et al. (2019) identified low efficiency of O₂ transmission as the source of longer processing times between the start and finish of the decomposition process and generation of high levels of GHG emissions. Their research utilized bench-top aerobic bioreactors, with rice hulls, sewage sludge and poultry manure as the feedstock with a 2VoltDC applied to the compost. The outcome was that levels of N₂O and CH₄ were reduced with an electric field compared to the control treatment but an increase in CO₂, 86.69 g CO₂ kg⁻¹ DM d⁻¹ and 55.01 g CO₂ kg⁻¹ DM d⁻¹, for the treatment and control, respectively, occurred during the peak CO₂ output on day 4.

4.2. CO₂ generation in field-based composting systems

In a comprehensive meta-analysis of 88 studies focused on emissions from dairy manure composting (silos, static, turning and windrow), Ba et al. (2020) reported CO₂ emissions from silos ranging from 2.35 to 206.68 g CO₂ kg (DM)⁻¹, from static systems ranging from -0.16 to 318.06 g CO₂ kg (DM)⁻¹, from turning systems ranging from 0.47 to 569.49 g CO₂ kg (DM)⁻¹, and from windrow systems ranging from 0.001 to 270.49 g CO₂ kg (DM)⁻¹ (Fig. 4). Clemens and Cuhls (2003), in a study of four different large-scale aerobic MSW treatment facilities,

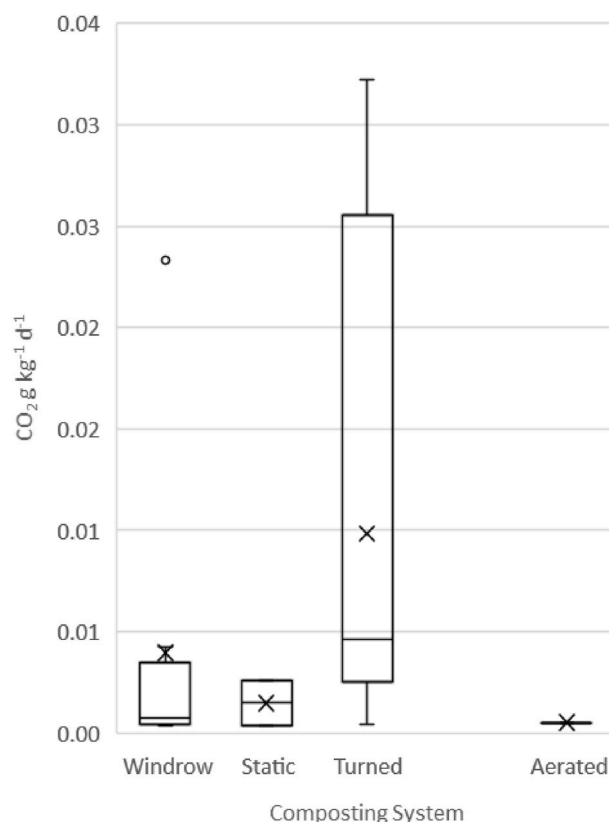


Fig. 4. Range of CO₂ emissions from different composting systems (data collated for this study from bio-waste, green waste, MSW and non-dairy manure)(adapted from Ba et al., 2020). *Whiskers denote the minimum and maximum values, "X" represents the mean, box lines denote (from the bottom) the first quartile, median and the third quartile, "o" denotes outliers.

found that CO₂ production levels ranged from 12 to 185 kg Mg⁻¹ of organic matter. They identified that the ranges in CO₂ emissions were influenced by aeration practices, with inefficient aeration causing lower CO₂ and NH₄ emission levels and higher levels of CH₄. Amlinger et al. (2008) conducted a literature review on CO₂ emissions from different composting systems (Amlinger et al., 2008) and identified ranges of 25–383 kg CO₂ Mg⁻¹ of fresh organic matter. Across all composting methods, operational influences and external variables resulted in the high range in gas emissions (Amlinger et al., 2008). Moreover, Amlinger et al. (2008) conducted several trials evaluating GHG emissions (CO₂, CH₄, NH₃ and N₂O), comparing a windrow-based system with three different substrates: green waste and sewage sludge and a backyard composting system using garden waste. They reported a range between 43 and 115 kg CO₂ Mg⁻¹ of fresh material (bio-waste), 118–194 kg CO₂ Mg⁻¹ of fresh material (garden waste) and 10–25 kg CO₂ Mg⁻¹ of fresh material (sewage sludge). Lou and Nair (2009) evaluated GHG emissions from landfilling and composting and concluded that the majority of compost research has focused on improving production processes, factors relating to the quality of the finished product, and methods for improving quality but very little research has focused on process emissions (Lou and Nair, 2009). Over the period between 2010 and 2019, there have been a range of studies evaluating total GHG emissions from composting systems to quantify emissions for use in Life Cycle Analysis, (Ahn et al., 2011; Gabriel et al., 2015; Jiang et al., 2011; Luo et al., 2014; Sánchez-Monedero et al., 2010).

In Fig. 4, the range of CO₂ emissions from four different composting systems illustrates that emissions are higher in turned composting systems. This aligns with research conducted by Ahn et al. (2011) a turned pile method with multiple pile turnings over an 80 day period. They found higher GHG emissions, including CO₂, occurred on days 10, 23,

36, and 52 compared to the non-turned static piles (Ahn et al., 2011). In a follow-up study, Mulbry and Ahn (2014) reported that dairy manure compost piles that were turned regularly generated greater amounts of CO₂ than static compost piles (Mulbry and Ahn, 2014). Ba et al. (2020) also reported cumulative CO₂ emission results from 27 studies, using dairy manure, in trials ranging from 18 to 168 days and identified turned composting systems as generating the highest emissions (Fig. 5).

4.3. Peak CO₂ emissions

CO₂ emissions from composting systems are directly correlated to the temperature increase within the system. As microbial activity increases, compost temperature goes up resulting from released metabolic heat during decomposition. Han et al. (2018) identified peak CO₂ concentrations at the transition point of the system from the mesophilic phase to the thermophilic phase, correlating to the increase in microbial activity within the pile (Han et al., 2018). This transitional phase occurs as mesophilic microbial groups die releasing their labile components to the system and specialized thermophilic bacteria and fungi become dominant and take over metabolic roles for decomposition of more recalcitrant carbon forms (Antunes et al., 2016). Peak CO₂ emissions typically occur between 2 and 12 days after mixing the feedstocks together, which aligns with the time frame for transition between the mesophilic phase to a thermophilic phase of composting. A review of 41 individual experiments over 15 separate studies (Fig. 6) on both open and closed composting systems, peak CO₂ emissions occurred early in the composting process, with an average of 7.6 days in open systems and an average of 5.2 days in closed systems (Table 4).

Extracting CO₂ from an in-vessel composting system when the system reaches peak or near peak CO₂ emissions would be advantageous to maximize available gas for enrichment practises in CEA or vertical farm systems. This is analogous to the operation of continuous anaerobic

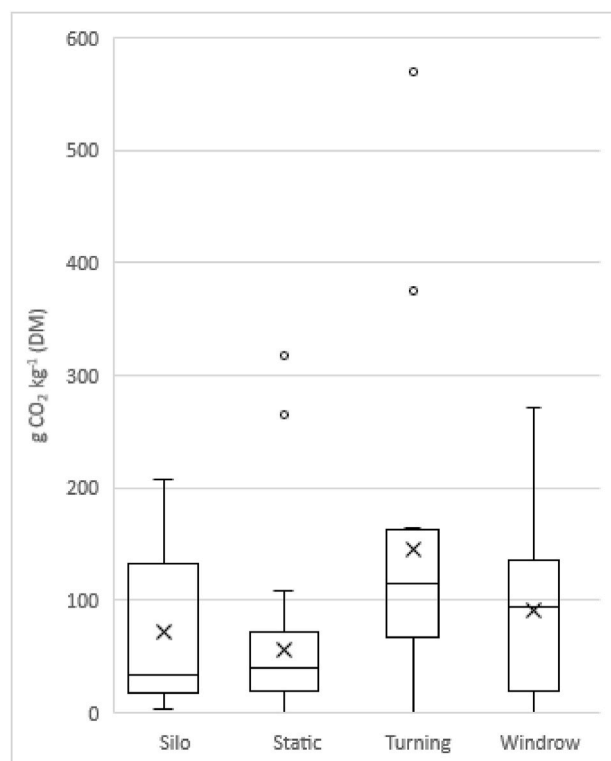


Fig. 5. Ranges of cumulative CO₂ emissions (g CO₂ kg⁻¹ (DM)) from dairy manure composting systems (Adapted from Ba et al., 2020). *Whiskers denote the minimum and maximum values, "X" represents the mean, box lines denote (from the bottom) the first quartile, median and the third quartile, "o" denotes outliers.

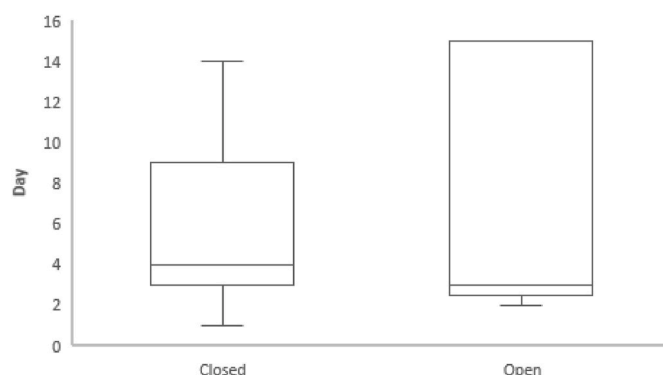


Fig. 6. Range of days when peak CO₂ concentrations were achieved in selected closed and open composting systems. Open refers to windrow and pile-based composting systems. Closed systems refer to in-vessel and lab-scale closed-container/vessel-based systems. *Whiskers denote the minimum and maximum values, "X" represents the mean, box lines denote (from the bottom) the first quartile, median and the third quartile, "o" denotes outliers.

digestion systems for biogas extraction (Hilkia et al., 2009) or the continuous operation of an algae bio-reactor (Croughan et al., 2015). Under continuous flow composting systems, the aim is to maintain constant operation of the system and receive a continuous output (Croughan et al., 2015). Batch systems aim to process a defined quantity of material that may be generated at specific times, i.e. crop harvest, rather than materials that are generated daily, i.e., manures or MSW organics. Vergara and Silver (2019) monitored GHG emissions continuously from a windrow compost of animal manure, reporting 50% of the cumulative CO₂ emissions over the first 21 days but steadily declining until the end of the 91 day study. Currently there are limited available studies on continuous CO₂ monitoring from continuous in-vessel composting systems.

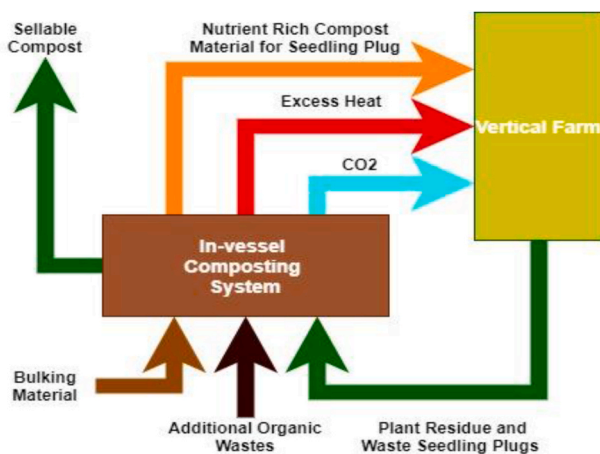
5. Application of in-vessel composting system to a vertical farming production framework

Closed bioreactor or contained in-vessel systems will have different CO₂ distribution patterns from open windrow or static pile systems. Windrow systems may be turned on a weekly or bi-weekly basis depending on height, width, porosity, heat profile, and moisture content, creating a variation in CO₂ distribution in the system and off-gassing from the central column. In-vessel systems are turned on a more frequent basis, creating an even distribution of microbial activity, and more consistent rates of CO₂ generation. Integration of in-vessel and other alternative systems with vertical farming is currently being demonstrated through the use of aquaponics; whereby wastewater from land-based aquaculture systems provide a nutrient solution for plant growth in CEA systems (Khandaker and Kotzen, 2018; König et al., 2016; Palm et al., 2018). Weidner et al. (2019) review the current knowledge, gaps, and outlooks for urban agriculture and argued for integration of decentralized anaerobic digestion and vermicomposting to address resource circularity in an urban production system. The current paper proposes the idea of resource circularity in vertical farming, similar to land-based aquaculture-vertical farming model, using in-vessel-composting bioreactors to add new value layers to the production system (Fig. 7).

The outputs of in-vessel composting systems including, CO₂, heat, and stabilized compost product, have the potential to be integrated within vertical farming systems. Utilizing CO₂ from compost processing to enrich plant production environments, heat to reduce energy consumption, and compost product as seedling plug stocks or nursery media all have potential to reduce operating costs and improve circularity. Excess compost can be sold on the bulk or retail market to generate extra revenue.

Table 4Peak CO₂ emissions from lab-based closed composting systems with different manures and sewage sludge.

g CO ₂ -C kg ⁻¹ d ⁻¹	Trial Days	Peak CO ₂ day	Compost Material	Bulking Agent	System	Reference
15.00	60	4	Chicken Manure, Sewage Sludge and Mature Compost	Rice Hulls	200L (0.2m ³) Cylinder with 14 Days intermittent Forced Aeration and 16 Days no Aeration	Tang et al. (2019)
23.65	60	4	Chicken Manure, Sewage Sludge and Mature Compost	Rice Hulls	200L (0.2m ³) Cylinder incorporating electric-field assistance with 14 Days intermittent Forced Aeration and 16 Days no Aeration	
429.27	107	14	Chicken Manure	Sawdust	150L Chamber with forced Aeration	Hwang et al. (2020)
337.91	107	7	Chicken and Cow Manure	Sawdust	150L Chamber with forced Aeration	
377.73	107	77	Chicken and Swine Manure	Sawdust	150L Chamber with forced Aeration	
277.09	107	7	Chicken Manure, plant residue and Spent Medium	Sawdust	150L Chamber with forced Aeration	
0.32	56	3	Dewatered Sewage Sludge with Zeolite	Wheat Straw	130L Cylinder with daily aeration	Wang et al. (2018)
0.33	56	3	Dewatered Sewage Sludge with CA-bentonite	Wheat Straw	130L Cylinder with daily aeration	
0.32	56	3	Dewatered Sewage Sludge with Medical Stone	Wheat Straw	130L Cylinder with daily aeration	
30.24	60	5	Sewage Sludge Using Electro-dewatering	Sawdust	40L Cylinder with forced Aeration	Wang et al. (2018)
8.79	60	6	Sewage Sludge using Mechanical dewatering	Sawdust	40L Cylinder with forced Aeration	
21.27	15	2	Sewage Sludge with low O ₂ conditions	Straw	150L Chamber with forced Aeration	He e al. (2018)
6.95	15	2	Sewage Sludge with high O ₂ Conditions	Straw	150L Chamber with forced Aeration	
1.46	15	2	Sewage Sludge with Ambient O ₂ Conditions	Straw	150L Chamber with forced Aeration	
3.28	36	9	Pig Manure with Medical Stone	Sawdust	6.2L (0.0062m ³) Cylinder with daily ventilation	Mao et al. (2019)
2.84	36	9	Pig Manure with Zeolite	Sawdust	6.2L (0.0062m ³) Cylinder with daily ventilation	
3.12	36	9	Pig Manure with Bamboo Biochar	Sawdust	6.2L (0.0062m ³) Cylinder with daily ventilation	
3.65	36	9	Pig Manure with Wood Vinegar	Sawdust	6.2L (0.0062m ³) Cylinder with daily ventilation	
14.45	53	4	Swine Manure	Sawdust	20L Cylinder with Forced Aeration	Chen et al. (2020)
11.97	65	4	Swine Manure with sulfamethoxazole	Sawdust	20L Cylinder with Forced Aeration	
13.04	65	4	Swine Manure with Norfloxacin	Sawdust	20L Cylinder with Forced Aeration	
14.21	15	4	Swine Manure with sulfamethoxazole and Norfloxacin	Sawdust	20L Cylinder with Forced Aeration	He et al. (2018)

**Fig. 7.** Integration pathway for in-vessel composting - CEA/Vertical Farming Systems.

Harris and Kountouris (2020), identified several drawbacks to vertical farming including energy usage from HVAC systems to modulate internal environments. In contrast, excess heat from a composting system could be used in pre-heating hot water supplies (similar to the use of solar thermal panels), as part of the air-exchange system, or even to increase the external air temperature around the outdoor condensing unit. Harris and Kountouris (2020), further state that benefits of vertical farming include ‘reduced inputs’ and development of a closed-loop system. However, waste plant material and spent seedling-plugs are not typically included in models for vertical farming systems, and as

such are not truly a closed-loop system. Dsouza et al. (2021), framed out a conceptual model for closed-loop CEA process that included on-site composting of plant residues to produce a compost for use as nutrient-rich seedling plug stock and gas for use internally. Carbon dioxide, as a by-product of composting could replace liquid CO₂, propane, or natural gas, generating a significantly lower carbon footprint. By increasing layers of valorization within a CEA or vertical farm system, the outcome is to reduce operating expenditures, another cited drawback to vertical farming (Harris and Kountouris, 2020). Currently, there is no existing literature or bodies of work demonstrating the integration of in-vessel composting with vertical farming or CEA; however, given the research underway with aquaponics-vertical farming models there is value in pursuing this concept.

5.1. Elevated CO₂ environments for plant production

The concentration of CO₂, as measured at the mid-troposphere (5 km from surface level), is 415 ppm (global average) (Change, N.G.C., n.d.), with surface level atmospheric CO₂ concentrations varying both spatially and temporally. Garc a et al. (2008) monitored CO₂ levels on a continuous basis for two years in an upper Spanish plateau, finding fluctuations diurnally and seasonally with reduced levels of CO₂ during the summer period. The concentration of CO₂ in different geographical areas is impacted by localized CO₂ producing systems ranging from industrial operations, road traffic, and composting facilities. These sources are balanced by CO₂ sinks including forestry cover, agricultural lands, green spaces and water bodies.

Carbon dioxide is an integral input component of photosynthesis. A number of environmental conditions influence the utilization rate of CO₂ including day length, temperature, and humidity, as well as the availability of key resources including nutrients, water, and critically,

CO₂ (Esmeijer and Bloem, 1999; Poudel and Dunn, 2017). Atmospheric levels of CO₂ are a limiting factor to plant growth, with plant CO₂ assimilation occurring during daylight hours when stomatal pores are open and gas exchange can occur. Even partially enclosed plant production environments, in the absence of CO₂ supplementation and under optimal growth conditions, can experience a decrease in CO₂ levels as photosynthetic rates outpace restricted diffusion of ambient CO₂. Research into CO₂ enrichment of greenhouses fluctuated repeatedly between the 1950's and 1980's (Mortensen, 1987) before more widespread adoption and application occurred across Europe and the USA (Li et al., 2020). The use of CO₂, through an enriched growing environment, is a method used by growers to improve yields through increased photosynthetic rates (Blom et al., 2002; Esmeijer and Bloem, 1999; Poudel and Dunn, 2017; Pratt et al., 2011).

Traditionally, in controlled environment agriculture, ventilation is a key operational parameter used to: control temperature, manage humidity levels and prevent condensation, and replenish CO₂. During the dark period, CO₂ levels increase due to plant and microbial respiration, which is not offset by light dependent photosynthesis reactions (Boulard et al., 1996; Iddio et al., 2020). In a study conducted by Frantz (2011), CO₂ levels in a closed environment were recorded as low as 175 ppm due to plant drawdown from photosynthesis. Similarly, Poudel and Dunn (2017) reported concentration ranges of 150–200 ppm during day-time hours in sealed greenhouses, highlighting the importance of ventilation in CO₂ replenishment. The addition of supplemental CO₂ on plant growth and crop responses in greenhouses is well documented in the literature, from Mortensen's substantial review (Mortensen, 1987) to Idso and Idso's ten year review of *Plant responses to atmospheric CO₂ enrichment in the face of environmental constraints: a review* (Idso and Idso, 1994). Blom et al. (2002) state that between 800 ppm and 1000 ppm is ideal for crops raised from seedlings including tomatoes, cucumbers, peppers and lettuce. At higher CO₂ levels (e.g., >1000 ppm), there are diminishing returns as saturation can be reached depending on available light flux.

Sources of CO₂ for greenhouses and CEA, include: a) complete air turnover through effective ventilation, b) combustion of biomass and the capture and re-direction of flue gas, c) dedicated natural gas/propane burners, or d) from liquid/compressed CO₂ tanks. Propane and natural gas are commonly combusted to produce CO₂, providing approximately a 1:3 ratio of kilograms of hydrocarbon fuel source to CO₂. Burners are placed directly in the growing space (greenhouse or vertical farm) and regulation of CO₂ can be controlled through sensors and ventilation. For larger greenhouse systems or multiple greenhouses, CO₂ enrichment can be achieved using a gas engine (e.g. boiler) or Combined Heat and Power system with an integrated flue gas condenser (Blom et al., 2002; Esmeijer and Bloem, 1999; Poudel and Dunn, 2017; Pratt et al., 2011). By-products from the combustion of fossil-fuel derived CO₂ also include heat and water. The production of heat in the winter-months can be viewed as a positive aspect to production, extending growing seasons in Northern climates but in the summer months heat removal becomes an additional cost to HVAC systems or other types of venting. Water production from these systems can also be controlled through increasing ventilation rates (Blom et al., 2002; Poudel and Dunn, 2017; Pratt et al., 2011). In contrast, liquid CO₂ removes the generation of heat and moisture by-products but increases the cost of the CO₂ source. Liquid CO₂ is more expensive per kg of gas compared to propane and natural gas (Blom et al., 2002; Poudel and Dunn, 2017; Pratt et al., 2011). Delivering CO₂ to plants can involve either filling the growing space with elevated CO₂ levels using a series of vents and ducts or delivering CO₂ at plant level in a targeted manner using a network of tubing placed within the plant crop canopy. A targeted approach to CO₂ elevation using an in-canopy approach increases efficiency of gas uptake and ensures a more even distribution of CO₂ uptake within the crop (Blom et al., 2002; Esmeijer and Bloem, 1999; Pratt et al., 2011). Further, delivery of CO₂ directly to the canopy aids in the management of water vapour/humidity within the canopy.

In a review of technical reports on the use of CO₂ enrichment, Nederhoff (1990) focused on enrichment using flue gas capture/direct combustion using burners and liquid CO₂. Other reports have evaluated technical and economic conditions of using flue gas capture and direct combustion using burners (Esmeijer and Bloem, 1999). The common theme in these reports is the use of combustion for the production of CO₂, with Pratt et al. (2011) stating that economically, the generation of CO₂ from composting does not compare economically to other forms of CO₂ generation. They provided an estimate of CAD\$170 t⁻¹ CO₂ using composting methods compared to CAD\$110 t⁻¹ CO₂ using a natural gas boiler system and capturing the flue gas. The costs were conditional on the existing cost of straw and natural gas. Moreover, the report did not take into account potential costs of waste disposal or carbon use from fossil fuels. Oklahoma State University Extension Service, produced a factsheet on CO₂ supplementation (Poudel and Dunn, 2017) briefly discussing the use of decomposition and fermentation methods for CO₂ generation, referencing several commercial options including the "CO₂ Boost Bucket", "Pro CO₂" and Exhale" mushroom bag, all of which are small scale and designed with the residential greenhouse market in mind. Poudel and Dunn (2017) highlighted the possibility of using decomposing organic matter in buckets to provide CO₂ but identified the need for significant amounts of substrate and physical space for storage.

5.2. Composting as a source of CO₂ enrichment of plant production environments

Conceptually, the use of compost to produce CO₂ for plant growth is a recognised option but has not been attempted at a commercial level and only a small number of studies have evaluated this option. Concerns from commercial operations relate to: a) uncertainty of cost, potential introduction of odours, pests, or pathogens, b) unwanted gases from the composting process, c) excess moisture or heat, d) leachate management, e) requirements for additional space, and f) management of the composting process (Esmeijer and Bloem, 1999; Province of Manitoba, 2021). There are a limited number of studies that have recently evaluated the potential of CO₂ production from compost in a plant production environment. Karim et al. (2020) evaluated CO₂ enrichment using crop residues and animal manure (CRAM) using five 0.34 m³ vessels in a 753 m³ sized greenhouse. They demonstrated both a sustained increase in CO₂ levels and an increase in plant productivity, with greater crop yields of tomatoes compared to the control greenhouse system. Hao et al. (2004) conducted a similar study finding a sustained increase in CO₂ and an increase in tomato yields compared to the ambient control.

5.3. Challenges of using CO₂ derived from composting systems

Technical barriers need to be overcome in order to transition CEA and vertical farm systems into a recapture and recycling model for their organic wastes. In particular, composting systems generate a gas mixture that may include components, such as methane, ethylene, ammonia, which are harmful to plant growth and development. Moreover, CEA production systems must meet different regulatory criteria with respect to biosecurity, pathogen management, and hygiene that field-based systems are not bound by. This will mean additional research on management of bioaerosols, particulate matter, pathogens, and handling of organic waste materials during the composting process to recapture and recycle CO₂. Each of these technical barriers has the potential to adversely impact plant production, human health, and operations of CEAs.

5.3.1. Particulate matter and bioaerosols

Particulate matter and bioaerosols, including microorganisms, microbial metabolites, fragments, and other microbial by-products from large-scale open composting systems have been documented as potential health hazards for compost system operators and communities surrounding compost facilities (Pearson et al., 2015; Wéry, 2014). In-vessel

composting systems have greater capacity to reduce dispersion of bioaerosols and particulate matter through the use of physical and biofilters. Air flow from in-vessel composting systems are directed through dedicated exhaust ports and often routed through biofilters to limit odour, particulate matter, and bioaerosols (Engineered Compost Systems, 2019; Green Mountain Technologies, n.d.; Brome Compost Inc., 2019; Renewable Carbon Management LLC, 2019; Global Composting Solutions Ltd., n.d.). The composting process, when achieving thermophilic temperatures (40 °C–60 °C) sustained for a minimum of three days, can reduce pathogen counts significantly (Mao et al., 2021). Pathogens including *E.coli* and *S. enterica* are commonly found in poultry manure and dairy/beef manure (Gutierrez and Schneider, 2020) but are eliminated (rendered inactive) through sustained exposure to thermophilic temperatures. However, pathogens including *L. pneumophila* and *P. aeruginosa* have been identified as opportunistic airborne pathogens that can enter via the natural environment (Loh and Soni, 2020; Mao et al., 2021) and are considered resistant to heat (Furukawa et al., 2018; Gutierrez and Schneider, 2020). In-vessel composting systems also have reduced avenues for contamination due to enclosure requirements and allows for greater process control to be administered over decomposition.

Secondary filtration can be incorporated to these systems, including the use of high efficiency particulate filters (HEPA) prior to any gas stream entering the CEA system.

5.3.2. Moisture removal

Moisture removal from composting gas streams are achieved through a condensation trap, whereby the output gas is passed through a heat exchange. This is a method currently built into in-vessel composting systems including those by Green Mountain Technologies, Brome Compost Inc., and Global Composting Solutions Ltd. A condensation trap method may not be sufficient for all CEA systems requiring a secondary process of solid desiccant dehydration. This process would use common absorbents including silica gel, activated alumina, or activated carbon in a column trap to remove excess moisture (Speight, 2017; Stewart, 2014). The disadvantage of this approach is a requirement for periodic maintenance of the absorbent via drying to recharge and cost.

5.3.3. Compost gas mixtures

Gas outputs from composting, while predominantly comprising of CO₂ can also contain ammonia (NH₃), methane (CH₄), and nitrous oxide (N₂O) (Ma et al., 2020). NH₃, CH₄ and N₂O emissions cannot be directly eliminated but can concentrations can be reduced through management of air flow, feedstock selection, adjusting process or retention time, and the use of filtration or traps. There are a range of different methods currently used for removing CO₂ from gas streams, particularly biogas from anaerobic digestion and natural gas. These methods include the use of water scrubbing, amine scrubbing, pressure swing adsorption and membrane filtrations (Petersson, 2013; Sun et al., 2015; Wasajja et al., 2020). Both water scrubbing and amine scrubbing operate under the principle that CO₂ is absorbed by a liquid (CO₂ chemically reacting with the scrubbing solvent or CO₂ dissolving into water) and separated from the desired gas of CH₄ before being stripped from the solvent/water (Yang and Li, 2014). Pressure swing adsorption (PSA) and membrane filtration are dry systems. PSA systems use a series of vessels operating under pressure and uses an adsorptive media to separate out the gas, (CO₂, CH₄) CO₂ is preferentially adsorbed due to smaller molecular size compared to CH₄, allowing CH₄ to be removed. After CH₄ has been removed, the system is depressurised allowing the captured gas to be released (CO₂). Membrane filtration requires the parent gas, under pressure, to be streamed through a single or a series of polymeric membranes to separate out CO₂ from CH₄. This occurs due to CO₂ being an ionically charged small molecule, allowing the CH₄ to pass through the membrane, creating two separate gas streams (Greene, 2018; Yang and Li, 2014).

6. Comparison of CO₂ enrichment approaches for plant production

To compare the cost of CO₂ produced via an in-vessel composting system against conventional methods of CO₂ enrichment, two hypothetical production CEA scenarios are presented below. The scenarios were based on two different fixed volumes for the growing spaces, with a fixed production of lettuce per m² per week, a fixed day length period, fixed ventilation, and a fixed target CO₂ requirement to achieve a concentration of 1000 ppm. Quantities (CO₂-C g kg⁻¹ d⁻¹) and Cost were calculated for liquid CO₂, Natural Gas, and Propane. Based on the calculated quantities of CO₂ and using data obtained from the literature review of peak CO₂ production per kg of material within a closed system, kilograms of compostable material to achieve a comparable quantity of CO₂ (to that of liquid CO₂) were calculated.

Determining the quantity of CO₂ for a given environment was based on a defined production level expressed as:

$$CO_2 \text{ Required} = \frac{\left[A_{Exch} \cdot \left[V_{Sys} \cdot \frac{PPM_{Des} - PPM_{Amb}}{1000000} \right] \cdot [NPP_{Total} \cdot CO_2C \cdot 0.545] \right]}{0.545} \quad (1)$$

where CO₂ Required (kg day⁻¹) provides the number of kg of CO₂ required for an enriched environment at a given target enrichment. A_{Exch} (m³ day⁻¹) represents the desired air changes within the system, V_{Sys} (m³) represents the growing chamber volume, PPM_{Amb} and PPM_{Des} (PPM) represent the CO₂ levels at ambient and desired levels, respectively, NPP_{Total} (kg day⁻¹) shows the sum of the total Net Primary Productivity (carbon content) for a given quantity of leafy greens, herbs and/or woody herbs, CO_2C represents the ratio of CO₂ to Carbon, V_{loss} represents percentage of uncontrolled air loss in the system, and 0.545681257 is the conversion factor from m³ of liquid CO₂ to kg of liquid CO₂.

The air exchange rate (A_{exch} (m³ day⁻¹)) was calculated using:

$$A_{exch} = \frac{Dl \text{ (hrs)}}{Dh_{a-exc}} \quad (2)$$

where Dl refers to the duration of Photosynthetic Active Radiation PAR in a given day and Dh_{a-exc} refers to the desired complete hourly air exchange for the growing chamber.

The volume of the growing chamber (V_{Sys} (m³)) has been calculated using the system's length (L_{Sys}), width (W_{Sys}) and height (H_{Sys}):

$$V_{Sys} = L_{Sys} \cdot W_{Sys} \cdot H_{Sys} \quad (3)$$

The Net Primary Productivity has been calculated through the following (adapted from Jenkins et al. (2003):

$$NPP_{Total} = [lg_y \cdot lg_{gf} \cdot lg_{cc}] + [hb_y \cdot hb_{gf} \cdot hv_{cc}] + [wh_y \cdot wh_{gf} \cdot wh_{cc}] \quad (4)$$

where NPP_{Total} (kg day⁻¹) is the sum of carbon content by weight of the different plants present in the model: leafy greens (NPP_{lg}), herbs (NPP_{hb}) and woody herbs (NPP_{wh}). NPP is calculated through identifying percentage carbon content lg_{cc} , hb_{cc} , and wh_{cc} (set at 49%, 46% and 45%, respectively and accounting for harvestable material); this can also be calculated through NPP equaling Gross Net Primary Productivity minus Respiration.

Quantities of CO₂ from natural gas and propane required have been derived from CO₂ Required and using the following conversion values:

Natural gas (kg day⁻¹):

$$NG_{Required} = CO_2 \text{ Required} \cdot 2.7026 \quad (5)$$

Propane (kg day⁻¹):

$$Ppn_{Required} = CO_2 \text{ Required} \cdot 2.9952 \quad (6)$$

For natural gas, 2.7026 is the conversion factor used to obtain an equivalent quantity of natural gas in kg day^{-1} to liquid CO_2 in kg day^{-1} , and 2.99528 is the conversion factor used to obtain an equivalent quantity of propane in kg day^{-1} to liquid CO_2 in kg day^{-1} . CO_2 -C ($\text{g CO}_2\text{-C d}^{-1}$) equivalences can be evaluated through multiplying each quantity of CO_2 source by their respective carbon proportional content: natural gas (CH_4): 75%, propane (C_3H_8): 82%, and liquid CO_2 : 27%. Carbon content is obtained through dividing the carbon content of each gas by its molecular weight. CO_2 from the in-vessel composting system is assumed to have a carbon content of 27% equivalent to that of liquid CO_2 (27%) (Brander and Davis, 2012).

6.1. Hypothetical examples of two scales of vertical farm production and CO_2 supply from composting and conventional sources

Basic Assumptions.

1. The scenarios and calculations are designed to present estimated CO_2 consumption and comparable costs; this paper was not intended to perform a full economic analysis of the use of a commercial scale in-vessel composting system to supply CO_2 to a CEA system. It should be noted that CEA system are integrated systems, for example, the use of propane as a CO_2 source contributes towards heat and moisture content, which may impact heat pump running costs (decreased costs in the winter and increased costs in the summer) and increase the need for air-exchanges or dehumidification to maintain an appropriate relative humidity.
2. The scenarios and calculations do not estimate the wider financial trade-offs/benefits with using in-vessel composting compared to traditional CO_2 enriching methods; these are discussed in general terms.
3. The scenarios and the selected 'container' have not been analysed for running costs, building envelope factors (U-value), installed HVAC systems, or lighting.
4. The scenarios and calculations do not account for a change in heat or moisture supply from the CO_2 sources and the costs associated with these.
5. The scenarios and calculations do not include potential costs savings associated with the sale of compost material or the use of composted material as a replacement for seedling plugs.
6. The scenarios assume filling the entire volume of the growing container and not using a targeted approach for CO_2 enrichment, i.e. delivering within the crop canopy using a duct and tubing system.
7. Table 5 provides the set parameters and assumptions for input factors.

Table 5
Assumptions for two scenarios of CO_2 generation from composting systems and supplementation into a CEA production facility.

Input	Value	Units	References
Ventilation Rate	9	Air changes per day	
Growing Period (daylight hours)	18	hours	
CO_2 Flow Rate	0.002	$\text{m}^3 \text{min}^{-1}$	
Ambient CO_2 Level	400	ppm	
Target CO_2 Level	1000	ppm	
Tiers	1–5		
Growing area of container	75	%	
Lettuce			
HarvestYear ⁻¹	10	–	Eaves and Eaves (2018)
YieldHarvest ⁻¹	5.34	Kg m^{-2}	Eaves and Eaves (2018)
Fuel Costs			
Liquid CO_2		$\text{\$ kg}^{-1}$	
Natural Gas		$\text{\$ kg}^{-1}$	
Propane		$\text{\$ kg}^{-1}$	

Grow Box Co., a UK and USA based company, produce a range of plug-and-play portable grow rooms and shipping container-based vertical farm solutions in a range of sizes and operating configurations (Grow Box Co, 2020). The standard dimensions offered are 3m, 6m or 12m in length by 2.4 wide and 2.5m in height, or 18 m^3 , 36 m^3 , and 72 m^3 , respectively. The modular containers are insulated, come with installed features including air-flow/air-exchange, a CO_2 enrichment system, HVAC, humidity control, and lighting. Growing space is offered through shelving tiers ranging from one to five tiers. Grow Box Co. has a target market demographic that includes small growers/home producers, cafes, restaurants, and hotels. Production is based on 5.34 kg m^{-2} of lettuce and the scenarios presented here calculate CO_2 requirements for a one and five tier production system. The dimensions of the Grow Box in this scenario are a 3.04m $\times$ 2.43m $\times$ 2.59m enclosed environment with an enrichment concentration of 1000 ppm CO_2

Table 6

Carbon content of composting feedstocks and mass of material required for CO_2 production equivalent to conventional gas sources for use in a modular vertical greenhouse system (Grow Box).

Carbon content $\text{g CO}_2\text{-C g kg}^{-1} \text{d}^{-1}$	$\text{kg}^{-1} \text{d}^{-1}$ (dry basis) to produce equivalent Carbon content for 1 Tier	$\text{kg}^{-1} \text{d}^{-1}$ (dry basis) to produce equivalent Carbon Content for 5 Tiers	Source
14.85	6.81	20.02	Tang et al. (2019) ^a
23.41	4.32	12.70	Tang et al. (2019) ^a
29.94	3.38	9.93	Wang et al. (2018) ^b
8.70	11.63	34.17	Wang et al. (2018) ^c
21.06	4.81	14.12	He et al. (2018) ^d
6.89	14.70	43.20	He et al. (2018) ^e
1.45	69.94	205.50	He et al. (2018) ^f
3.25	31.16	91.55	Mao et al. (2019) ^g
2.81	36.00	105.78	Mao et al. (2019) ^h
3.09	32.80	96.38	Mao et al. (2019) ⁱ
3.61	28.05	82.41	Mao et al. (2019) ^j
14.31	7.07	20.78	Chen et al. (2020) ^k
11.85	8.54	25.09	Chen et al. (2020) ^l
12.91	7.84	23.04	Chen et al. (2020) ^m
14.07	7.20	21.14	Chung et al. (2018) ⁿ

^a Poultry manure, sewage sludge and mature compost using rice hulls as a bulking agent.

^b Sewage sludge using electro-dewatering.

^c Sewage sludge using mechanical dewatering.

^d Sewage sludge with low O_2 conditions.

^e Sewage sludge with high O_2 conditions.

^f Sewage sludge with ambient O_2 conditions.

^g Swine manure with medical stone.

^h Swine manure with zeolite.

ⁱ Swine manure with bamboo biochar.

^j Swine manure with wood vinegar.

^k Swine manure.

^l Swine manure with sulfamethoxazole.

^m Swine manure with norfloxacin.

ⁿ Swine manure with sulfamethoxazole and norfloxacin.

(requiring an average supply of 101.2 g CO₂-C day⁻¹ for one tier and 297.4 g CO₂-C day⁻¹ for five tiers (Table 6). This would result in a production of 5.72 kg week⁻¹ for a one tier system up to 28.62 kg week⁻¹ for a five tier system.

Evaluation of several composting studies measuring peak CO₂ emissions (g CO₂ d⁻¹ kg⁻¹ of compost) from closed composting systems (Chen et al., 2020; He et al., 2018; Mao et al., 2019; Tang et al., 2019; Wang et al., 2018) were used to estimate the quantity of composting biomass required to achieve an equal daily carbon content (target of 101.21 g CO₂-C day⁻¹) as provided by natural gas, propane or liquid CO₂ in a one tier system. Using the above production target for lettuce and enriching the environment to 1000 ppm of CO₂, a range of between 3.38 and 69.94 kg (dry basis) of organic feedstock for an in-vessel composting system with one tier and 9.93–205 kg (dry basis) of material for a production system with five tiers would need to be composted on a daily basis (see Table 6).

Commercial continuous flow in-vessel composting systems typically process raw materials over 10–21 days to achieve a homogenized product. For the scale used in this scenario, one commercially available system, the Earth-cube (Green Mountain Technologies, 2021), could be used with a daily loading capacity of 22.7 kg day⁻¹ (~8 tonnes per year). Based on an average value of 18.2 kg day⁻¹ (dry basis) (Table 6, Tier 1), adjusted to 29.25 kg day⁻¹ on a wet basis to achieve a target moisture content of 60% MC to generate 101.21 g CO₂-C day⁻¹. To process this quantity of biomass, 1 to 2 Earth-cube systems would be required and this would increase to four systems for a five tier production system with a target of 297.41 g CO₂-C day⁻¹, requiring 85.9 kg day⁻¹ at 60% MC (31 tonnes per year). To put these values for a one or five tier production system into perspective (29.25 kg day⁻¹ to 85.9 kg day⁻¹), Filimonau et al. (2021) conducted a food waste study in a UK-based restaurant and estimated 14.75 tonnes of organic waste generated per year, equivalent to 40.4 kg of waste per day. Aamir et al. (2018) in a wide-ranging study of 58 eating establishments, found that daily quantities of waste from restaurants ranged from 16.2 kg day⁻¹ in low-end restaurants to 56.5 kg day⁻¹ in fine dining restaurants, with an average of 35.6 kg across all restaurant types. Roy and Ballantine (2020), conducted 59 interviews in Canada and New Zealand, finding an increased preference by restaurants and chefs to purchase produce locally due to price, consistency of product supply, and relationships with growers. These findings are similar to Murphy and Smith (2009) who interviewed 11 chefs in several Canadian provinces, finding that product quality, meeting supply deadlines, and addressing problems were key drivers for locally purchased produce. This model of local sourcing has taken a further step with some restaurants having space for on-site production of fresh ingredients, or ‘chef gardens’, especially for herbs, salad and vegetable produce. Recent articles (Durrani, 2016; Lerner, n.d.; Morrissey-Swan, 2018) cite a range of reasons for on-site production, both outdoors and greenhouse-based, including reducing carbon footprints, production of unusual items, and freshness of produce.

Calculated costs associated with elevating CO₂ to the 1000 ppm threshold for this scenario range from CAD\$4.07 to CAD\$11.00 day⁻¹ for liquid CO₂, CAD\$0.43 to CAD\$3.15 day⁻¹ for natural gas, and CAD\$0.18 to CAD\$1.49 day⁻¹ for propane (Table 7). Equipment costs also need to be included in the comparison of CO₂ sources. For instance, the

current cost for an Earth-cube composter is CAD\$2,000 per unit, with two small units needed for a one tier Grow Box vertical farm system producing 5.72 kg of lettuce per week. A five tier production system requires four composting systems at a cost of CAD\$8,000. The cost for a single propane burner required to enrich CO₂ concentrations in the production space, ranges between CAD\$900 to CAD\$1,100 (Indoor Growing Canada, 2021), plus tank rental costs, and approximately CAD\$97.99 yearly in propane costs in a one tier production system. At this production scale, a natural gas heating system with flue-gas extraction would not be viable with the cost for heating systems ranging from CAD\$7,000 to CAD\$10,000, in addition to the flue gas extraction set up.

A number of uncertainties exist with respect to the operational costs of extracting CO₂ from a composting system. For instance, a continuous flow in-vessel composter will have energy requirements associated with air circulation, i.e., blowers, mixing (rotating a drum or an internal shaft with paddles), pumps for gas circulation, air/gas filtration, moving feedstock into the system, and heat removal. Energy requirements for the composting system will vary according to the size of the production operation, i.e., quantity of biomass generated and feed-in rate, and type of composting system. For example, a 125W blower operating on a 24hr basis at a rate of \$0.174kwh⁻¹ would result in a daily running cost of \$0.53, comparable to Natural Gas. For a one-tier Grow Box production system, \$1.06 per day for aeration could be expected based on two composting units.

Integration of in-vessel composting-vertical farming systems has the potential to push the boundary of resource circularity, especially when applied to restaurants that are looking for local supplies of fresh produce. Fueling the in-vessel composting system with organic residues from the vertical farm (cutting down external waste disposal costs) has cost reduction opportunities and benefits for plant production. Long-term prospects from circularizing CO₂ enrichment may also include carbon credits and energy-offsets from reduced transportation of foods from local vertical farm production. Another valorization opportunity may be in using the compost products to extract additional fertility for nutrient solutions or converting the stabilized compost into a seedling plug stock alternative media, however, these are still areas requiring further research.

6.2. Future direction and recommendations

The capture of CO₂ from industrial processes including fertiliser plants (Haugen et al., 2017; Zhuang et al., 2012) is well documented, as is the technology for CO₂ scrubbing (Chung et al., 2018; Krótki et al., 2020; Kapoor et al., 2021). Composting systems can produce CO₂ emissions at levels that may meet requirements of small to medium sized plant production systems. Pilot studies evaluating integration of bioreactor systems with CEA facilities will be needed to determine how factors such as mixed gas scrubbing, bioaerosol filtering, and biosecurity of the production system can be optimized. As the concept of circular economies or closed loop systems grows, more industries and infrastructure will be built around resource integration and development of valorization opportunities (Malolan et al., 2021; Calone et al., 2019; Nuwansi et al., 2021). There is a clear lack of research focusing on the valorization and use of metabolic by-products from aerobic composting as part of a circularization strategy in CEA and vertical farm systems. Growing investments in CEA and vertical farming creates new synergies to process residual biomass, i.e., non-harvestable residues, using composting approaches for recapture and utilization of carbon that can be recycled back into plant production. With an increased desire for local produce by consumers, chefs, and restaurants there are clear opportunities for vertical farming systems to include in-vessel composting to circularize resources and reduce costs (Dukeshire et al., 2011; Murphy and Smith, 2009; Roy and Ballantine, 2020; Zhang et al., 2020). In-vessel composting systems are not only likely to become more integral and valuable in achieving successful waste management objectives; more crucially they are a key component of zero waste principles.

Table 7

CO₂ requirements per day for production at one and five tiers of an Earth-cube and costs associated with conventional sources of CO₂ gas.

CO ₂ Source	g d ⁻¹ from source		Carbon Content CO ₂ -C g d ⁻¹ from source		\$ d ⁻¹	
	1 Tier	5 Tiers	1 Tier	5 Tiers	1 Tier	5 Tiers
Liquid CO ₂	369.72	1086.38	99.82	293.32	4.07	11.00
Natural Gas	136.80	401.97	102.60	301.48	0.43	3.15
Propane	123.43	362.70	101.21	297.41	0.18	1.49

7. Conclusions

The aim of this review is to highlight the potential scope for capturing and recycling metabolic CO₂ gas from composting of residual biomass and organic wastes in CEA or vertical farm systems. Composting has generated significant research interest and outcomes over the past few decades as a waste management tool. This paper has identified some of the opportunities for CO₂ generation from composting systems to meet the needs of small-scale commercial CEA systems and vertical farms. The current gaps in knowledge that will need additional research include quantifying CO₂ from the non-edible biomass of different CEA systems and vertical farms for use in continuous feed-in compost systems, developing models of compositional analysis of compost gases, heat valorization and management, development of compost media as a seedling plug stock alternative, quantification of nutrient content from compost for reuse in hydroponic solutions, and operational life cycle analyses of CEA-composting in integrated systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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